AP CALCULUS AB

Unit 1 • Lesson 1-8 • Notes (filled in)

Lesson 1-8 • MVT

Notes (filled in)

Learning goals

- State the Mean Value Theorem and check its hypotheses.
- Find the value(s) of c guaranteed by the MVT.
- Apply the MVT to tables and contexts.

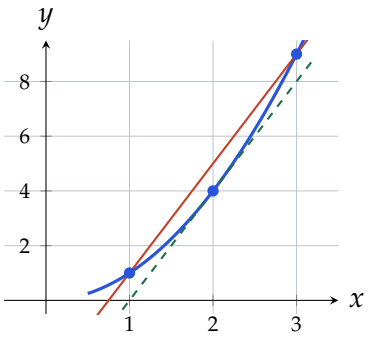
Vocabulary. Mean Value Theorem (MVT), differentiable, open interval, secant slope, tangent slope.

Mean Value Theorem

If f is __continuous__ on $[a,b]$ and __differentiable__ on (a,b), then there is at least one c in (a,b) such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

Some tangent line is __parallel__ to the secant line: at some instant, the instantaneous rate of change equals the average rate of change.



$f(x) = x^2$ on $[1,3]$: secant slope 4; the tangent at $c = 2$ has slope 4 too.

Example 1 • Finding c

Find the value(s) of c guaranteed by the MVT for (a) $f(x) = x^2$ on $[1,3]$; (b) $g(x) = x^3$ on $[0,3]$; (c) $k(x) = \frac{1}{x}$ on $[1,4]$. (Use $f'(x) = 2x$, $g'(x) = 3x^2$, $k'(x) = -\frac{1}{x^2}$ from Lesson 1-7.)

(a) $\frac{9-1}{2} = 4 = 2c$: $c = 2$.
(b) $\frac{27-0}{3} = 9 = 3c^2$: $c = \sqrt{3} \approx 1.732$ (the negative root is not in $(0,3)$).
(c) $\frac{\frac{1}{4}-1}{3} = -\frac{1}{4} = -\frac{1}{c^2}$: $c = 2$.

Example 2 • When the MVT does not apply

Explain why the MVT does not apply: (a) $f(x) = |x|$ on $[-1, 1]$; (b) $g(x) = \frac{1}{x}$ on $[-1, 1]$.

(a) Not differentiable at $x = 0$ (corner). The secant slope is 0, yet no tangent is horizontal.
(b) Not continuous on $[-1, 1]$ (undefined at 0).

Example 3 • A table in context

A car's position $s(t)$ (miles) is differentiable, with $s(0) = 0$ and $s(2) = 130$ (t in hours). Must the car's speed have been exactly 65 mph at some time?

s is differentiable, so continuous; the MVT applies on $[0, 2]$: $s'(c) = \frac{130-0}{2} = 65$ for some c in $(0, 2)$. Yes: at some instant the velocity was 65 mph.

Common mistakes

- Forgetting to check differentiability (corners, cusps) or continuity.
- Giving a c that is not inside the open interval.

Check your understanding

1. For $f(x) = x^2$ on $[0, 4]$, the MVT gives $c =$ __2__.
2. If $f(2) = f(5)$ and f satisfies the hypotheses, then $f'(c) =$ __0__ for some c in $(2, 5)$.